Domain 2 Master Cheat Sheet - Configuration Management and IaC (17% of DOP-C02)

One place for the facts that win Domain 2 questions: CloudFormation (core, change control, nesting, helper scripts, extending, rollback, StackSets), CDK, SAM, Systems Manager, AppConfig, Organizations, Control Tower, Service Catalog, Config, tagging, RAM and the extras (OpsWorks, automation with Lambda / Step Functions / Scheduler, CloudFormation Git sync and IaC generator). Read Part 1 first, then use the **keyword list (Part 20)** as a last-day review.

Part 1 - How Domain 2 questions work

The three task statements

Task	What the exam asks you to do
2.1 Define cloud infrastructure and reusable components	CloudFormation templates, parameters, nested stacks, modules, custom resources, macros, CDK constructs, SAM, change sets, drift, rollback, helper scripts
2.2 Automate multi-account / multi-Region setup	Organizations, OUs, SCPs, Control Tower, Account Factory / AFT / CfCT, StackSets, delegated admins, IAM Identity Center, RAM
2.3 Automate complex tasks at scale	Systems Manager (Run Command, State Manager, Patch Manager, Automation, Session Manager, Parameter Store), AppConfig, Config rules + remediation, Service Catalog, tagging

How to pick the right answer

1. **Find the requirement words**: least operational overhead, no manual steps, all accounts including new ones, prevent vs detect, without redeploying, least privilege.
2. **Prevent vs detect vs fix**: prevent = **SCP / IAM / proactive Hooks**; detect = **Config rules**; fix = **SSM Automation remediation**.
3. **Push vs pull**: push to every account = **StackSets / Control Tower / CfCT**; self-service pull = **Service Catalog**.
4. **Org-wide + new accounts automatically** = service-managed StackSets auto-deployment, org conformance packs, org aggregators, OU sharing in RAM, delegated admin with auto-enable.
5. **Keep people out of the management account** = **delegated administrator**.
6. **Eliminate answers with a wrong fact** (tag policy requiring tags, aggregator remediating, SCP affecting the management account, Config rule preventing changes).

Part 2 - CloudFormation core

Topic	Remember
Sections	`AWSTemplateFormatVersion`, `Description`, `Metadata`, `Parameters`, `Rules`, `Mappings`, `Conditions`, `Transform`, `Resources` **(only required)**, `Outputs`
Parameter types	String, Number, List, CommaDelimitedList, **AWS-specific** (`AWS::EC2::KeyPair::KeyName`, `AWS::EC2::VPC::Id`...), `AWS::SSM::Parameter::Value<...>` (latest AMI from public parameter)
Parameter options	`AllowedValues`, `AllowedPattern`, `Min/MaxLength`, `Default`, `NoEcho` (masks, not encryption)
Mappings	Static lookup with `Fn::FindInMap` (Region -> AMI, env -> size)
Conditions	`Fn::Equals`, `Fn::If`, `Fn::And`, `Fn::Or`, `Fn::Not`; `Condition:` on resources / outputs; `AWS::NoValue` to drop a property
Intrinsics	`Ref`, `Fn::GetAtt`, `Fn::Sub`, `Fn::Join`, `Fn::Split`, `Fn::Select`, `Fn::GetAZs`, `Fn::ImportValue`, `Fn::Base64`, `Fn::Cidr`, `Fn::Length`, `Fn::ToJsonString`, `Fn::ForEach` (LanguageExtensions)
Pseudo parameters	`AWS::Region`, `AWS::AccountId`, `AWS::StackName`, `AWS::StackId`, `AWS::Partition`, `AWS::URLSuffix`, `AWS::NoValue`, `AWS::NotificationARNs`
Outputs / exports	`Export: Name` -> `Fn::ImportValue` in another stack; **same account + Region**; export names unique; **can't delete / change an export in use**
Dynamic references	`{{resolve:ssm:name:version}}`, `{{resolve:ssm-secure:...}}` (supported properties), `{{resolve:secretsmanager:secret:SecretString:key}}` - secrets never in template
DependsOn	Explicit ordering when there's no `Ref` / `GetAtt` dependency
Service role	Stack uses a **CloudFormation service role** (`--role-arn`) so users need only `iam:PassRole`, not resource permissions

Part 3 - Change control

Feature	Remember
Change sets	Preview **Add / Modify / Remove** and **Replacement True / False / Conditional** before executing; required for templates with transforms (SAM)
Update behaviours	**No interruption**, **some interruption**, **replacement** (new physical ID - data loss risk)
Drift detection	Finds **manual changes** to supported resources (IN_SYNC / MODIFIED / DELETED / NOT_CHECKED); Config rule `cloudformation-stack-drift-detection-check`
Stack policy	JSON **Allow / Deny** on `Update:Modify`, `Update:Replace`, `Update:Delete`, `Update:*` for logical IDs; protects **all users** during updates; **temporary override** for one update
Termination protection	Prevents **stack deletion** (nested stacks inherit from root)
DeletionPolicy	**Delete** (default most), **Retain**, **Snapshot** (EBS, RDS, ElastiCache, Neptune, Redshift), **RetainExceptOnCreate**
UpdateReplacePolicy	Same values - what happens to the **old resource when replaced**
Resource import	Bring existing resources into a stack (needs DeletionPolicy in template)

Part 4 - Nested stacks, cross-stack, modules

Option	When
Nested stacks (`AWS::CloudFormation::Stack`)	**Reusable components** deployed **together**, lifecycle tied to parent; pass values with Parameters / `GetAtt Outputs.X`; beat the **500-resource** limit; update via **root** stack
Cross-stack references (Exports / ImportValue)	**Independent lifecycles** (network stack used by many app stacks); same account + Region; **export can't change while imported**
Modules (`...::MODULE`)	Reusable **resource bundle merged into the same stack** (no extra stack) from the Registry
SSM parameters	Loose coupling across accounts / Regions / tools (write output to Parameter Store, read with dynamic reference)

Part 5 - Helper scripts and policies

Item	Remember
cfn-init	Reads `AWS::CloudFormation::Init` metadata: packages, groups, users, sources, files, commands, services; **configSets** for order
cfn-signal	Sends success / failure to a **CreationPolicy** or **WaitCondition** (`cfn-signal -e $? --stack --resource --region`)
cfn-hup	Daemon that **detects metadata changes** (default every **15 min**) and runs hooks (re-run cfn-init)
CreationPolicy	Resource not CREATE_COMPLETE until **N signals** within **Timeout** (max 12h); ASG `MinSuccessfulInstancesPercent`
WaitCondition + handle	Wait for **external** signals / resources outside the instance; prefer CreationPolicy for EC2 / ASG
UpdatePolicy AutoScalingRollingUpdate	`MaxBatchSize`, `MinInstancesInService`, `PauseTime`, `WaitOnResourceSignals`, `SuspendProcesses`
AutoScalingReplacingUpdate	`WillReplace: true` = new ASG, swap after success (fast rollback)
AutoScalingScheduledAction	`IgnoreUnmodifiedGroupSizeProperties`
Others	`CodeDeployLambdaAliasUpdate` (SAM), ElastiCache `UseOnlineResharding`, OpenSearch `EnableVersionUpgrade`
Failure clue	**Timed out waiting for signals** = cfn-signal not sent (script failed, no outbound access, wrong resource name)

Part 6 - Extending CloudFormation

Feature	Remember
Custom resource	`Custom::X` with `ServiceToken` (Lambda / SNS); Create / Update / Delete; **must PUT response to the pre-signed S3 URL** or stack waits up to **1 hour**; **stable PhysicalResourceId**; **Delete must succeed**
Macros / transforms	Rewrite templates: `AWS::Serverless` (SAM), `AWS::Include` (S3 snippet), `AWS::LanguageExtensions` (ForEach), SecretsManager, CodeDeployBlueGreen; need `CAPABILITY_AUTO_EXPAND`; macro must exist in **same account + Region**
Registry	**Public** extensions (activate, third party) and **private** resource types (`cfn submit`), modules, hooks; roll out with StackSets
Hooks	**Proactive** checks **before** provisioning (Guard / Lambda); **FAIL** or **WARN**; Control Tower proactive controls
cfn-lint / cfn-guard	Shift-left template validation in the pipeline
Capabilities	`CAPABILITY_IAM`, `CAPABILITY_NAMED_IAM` (custom names), `CAPABILITY_AUTO_EXPAND` (macros / nested with transforms)

Part 7 - Rollback and troubleshooting

Status / tool	Action
ROLLBACK_COMPLETE (failed create)	Can't update: **delete and recreate** (pipeline `REPLACE_ON_FAILURE`)
UPDATE_ROLLBACK_COMPLETE	Healthy previous state - fix and update again
UPDATE_ROLLBACK_FAILED	`continue-update-rollback` (fix cause, or `--resources-to-skip`, nested = root + `Child.Resource`)
DELETE_FAILED	Fix dependency or delete with `--retain-resources`
Rollback triggers	Up to **5 CloudWatch / composite alarms**, **monitoring 0 to 180 min**
Disable rollback	`--disable-rollback` / `--on-failure DO_NOTHING` / preserve resources -> retry / `rollback-stack`
`--on-failure DELETE`	Remove failed create completely
`cancel-update-stack`	Stop an in-progress update (rolls back)
Root cause	**First (oldest) FAILED event**, child stack events, **CloudTrail** for AccessDenied
Alerts	**EventBridge** stack status change -> SNS

Part 8 - StackSets

Topic	Remember
Concept	One template -> **stack instances** in **many accounts x Regions** from an **admin account**
Self-managed	You create **AWSCloudFormationStackSetAdministrationRole** (admin) + **AWSCloudFormationStackSetExecutionRole** (targets); **any account** (incl. outside org, management account); **no auto-deploy**
Service-managed	**Organizations trusted access**, automatic roles, **OU targets**, **auto-deployment**, **delegated admin** (up to 5, `--call-as DELEGATED_ADMIN`); **never deploys to the management account**
Auto-deployment	Account joins OU = instances created; leaves = deleted or kept (**RetainStacksOnAccountRemoval**); **new Region = manual** `create-stack-instances`
Account filters	**DIFFERENCE** (exclude), INTERSECTION, UNION
Operation preferences	**RegionOrder**, SEQUENTIAL / PARALLEL, **MaxConcurrent**, **FailureTolerance (per Region)**, STRICT / SOFT; **managed execution** (no OperationInProgressException)
Lifecycle	Parameter overrides per instance; **delete instances before StackSet**; `--retain-stacks`; OUTDATED / INOPERABLE; drift detection
Trap	**IAM is global** - fixed-name roles in a multi-Region StackSet fail in the 2nd Region

Part 9 - AWS CDK

Topic	Remember
Model	Code (TS, Python, Java, C#, Go) -> **synthesizes CloudFormation** (`cdk.out`); App -> Stage -> **Stack** -> Construct (scope, id, props)
Constructs	**L1 = Cfn*** (raw), **L2** = curated defaults + `grant*`, **L3** = patterns (ALB Fargate service)
Escape hatch	`node.defaultChild` -> L1, `addPropertyOverride`
Bootstrap	`cdk bootstrap` per **account + Region** (CDKToolkit: S3, ECR, roles); cross-account `--trust <pipeline account>` + `--cloudformation-execution-policies`
CLI	**synth** (templates, no AWS change), **diff** (replacements, IAM changes), **deploy** (assets + change set, `--require-approval broadening` default), `--hotswap` / `watch` = **dev only**
Traps	**Logical ID = path + hash** (rename = replacement, `overrideLogicalId`); **RemovalPolicy RETAIN** default for stateful; **lookups need** `env`; **commit** `cdk.context.json`; Aspects / cdk-nag; `SecretValue`
CDK Pipelines	Source -> Synth -> **UpdatePipeline (self-mutation)** -> Assets -> Stages / `addWave` (parallel); `ManualApprovalStep`, post `ShellStep` + `envFromCfnOutputs`; first deploy manual; `crossAccountKeys: true`
Other	`cdk migrate` (CFN -> CDK), `cdk import` (adopt), `exportValue` (deadly embrace), `crossRegionReferences`

Part 10 - AWS SAM

Topic	Remember
Model	`Transform: AWS::Serverless-2016-10-31` (required); Function, Api, HttpApi, SimpleTable, LayerVersion, StateMachine, Application (SAR), Connector; **Globals**
Policies	**SAM policy templates** (`DynamoDBCrudPolicy`, `S3ReadPolicy`) / **Connectors** for least privilege
Safe deploys	**AutoPublishAlias + DeploymentPreference** (Canary10Percent5Minutes, Linear..., AllAtOnce, **Alarms**, Pre / PostTraffic hooks calling **PutLifecycleEventHookExecutionStatus**)
CLI	`sam build` (`--use-container` for native deps), `sam package` (= `aws cloudformation package`, S3 URIs), `sam deploy` (`--guided` -> samconfig.toml, **CAPABILITY_IAM + CAPABILITY_AUTO_EXPAND**, `--no-fail-on-empty-changeset`, `--config-env`), `sam sync --watch` (dev only), `sam pipeline init --bootstrap`
Local testing (**Docker**)	`sam local invoke` (one event), `start-api` (:3000), `start-lambda` (:3001 for SDK tests), `generate-event`; **no IAM enforcement, no service emulation**
Traps	S3 event bucket must be in the same template; SimpleTable = one key; local CodeUri must be packaged

Part 11 - Systems Manager

Capability	Remember
Managed node	**Agent +** `AmazonSSMManagedInstanceCore` **(or DHMC) + 443 to ssm / ssmmessages / ec2messages** (VPC endpoints in private subnets); hybrid = **activation** (`mi-`)
Run Command	**Once, now**; tags / resource groups; **concurrency + error threshold**; output **S3 / CloudWatch Logs** (console 2,500 chars)
State Manager	**Desired state on a schedule**; **new tagged instances included** (agent updates, CloudWatch agent, Inventory)
Patch Manager	**Baseline** (rules, auto-approve after N days, custom - predefined not editable) + `Patch Group` **tag** (one baseline per OS) + **maintenance window** running **AWS-RunPatchBaseline** (**Scan / Install**, RebootIfNeeded / NoReboot); org-wide **patch policies**
Inventory	`AWS-GatherSoftwareInventory`; **Resource Data Sync -> S3 -> Athena**
Session Manager	No SSH / bastion / inbound ports; IAM `ssm:StartSession` **+ tag conditions**; **logging to S3 / CloudWatch + KMS**; **port forwarding** (`ToRemoteHost` for private RDS)
Parameter Store	String / StringList / **SecureString (**`kms:Decrypt`**)**; **Standard 4 KB free** vs **Advanced 8 KB + policies + RAM sharing** (no downgrade); hierarchy, labels; **public AMI parameters**; rotation = Secrets Manager
Automation	Runbooks (`executeAwsApi`, `runCommand`, `executeScript`, `approve`, `branch`); **AutomationAssumeRole**; triggered by **Config remediation**, EventBridge, windows; `AWS-UpdateLinuxAmi`, `AWSSupport-ResetAccess`
Distributor / **Fleet Manager**	Package + install (with State Manager) / UI incl. **browser RDP (no 3389)**
Others	**Change Calendar** (freeze), OpsCenter, Change Manager, Quick Setup

Part 12 - AWS AppConfig

Topic	Remember
Purpose	**Dynamic config / feature flags without redeploying code**
Model	Application -> **Environment (CloudWatch alarm monitors)** -> **Configuration profile** (Feature flags = **hosted store only**; freeform from hosted, S3, Parameter Store, SSM doc, Secrets Manager, CodePipeline) -> Deployment (version + environment + strategy)
Validators	**Before deployment**: **JSON Schema** (structure) + **Lambda** (logic, needs invoke permission)
Strategies	AllAtOnce, Linear50PercentEvery30Seconds (test), **Canary10Percent20Minutes** (prod), Linear20PercentEvery6Minutes; custom: duration, linear / exponential, growth %, **final bake time**
Rollback	Alarm during rollout **or bake** = automatic rollback; `StopDeployment`; **one deployment per environment**
Retrieval	`StartConfigurationSession` **+** `GetLatestConfiguration` (empty = no change); **AppConfig Agent / Lambda extension (**`localhost:2772`**)**, ECS / EKS sidecar
Extras	Multi-variant flags (context); extensions (PRE_START_DEPLOYMENT, ON_DEPLOYMENT_ROLLED_BACK -> SNS / EventBridge); Evidently retiring

Part 13 - AWS Organizations

Topic	Remember
Structure	Management account (pays, **not restricted by SCPs**, no workloads) + OUs (nested **5**) + accounts; **all features** needed for policies
Accounts	Created = **OrganizationAccountAccessRole**; invited = create role manually
SCPs	**Never grant**; apply to member principals **incl. root**; not management account / service-linked roles; **explicit deny wins**, Allow needed at **every level**; deny-list (FullAWSAccess) vs allow-list
SCP patterns	Deny LeaveOrganization / disabling CloudTrail-GuardDuty-Config; **Region restriction with NotAction for global services** + `ArnNotLike aws:PrincipalARN`; require tags / encryption / IMDSv2
Other policies	**RCP** (resource data perimeter), **declarative** (EC2 baseline), **tag**, backup, AI opt-out
Condition keys	`aws:PrincipalOrgID` (resource policies, whole org), `aws:PrincipalOrgPaths`, `aws:ResourceOrgID`
Billing	One bill, volume discounts, **RI / SP sharing (turn off per account)**, one free tier, cost allocation tags in management account
Delegation	**Trusted access** (org CloudTrail, Config, GuardDuty, StackSets, RAM...); **delegated admin** = least use of management account; centralized root access; org events in **us-east-1**

Part 14 - AWS Control Tower

Topic	Remember
Landing zone	**Home Region fixed**, governed Regions + **Region deny**; **Security OU**: **Log Archive** (logs) + **Audit** (security, SNS, aggregator, delegated admin); org trail, Config, Identity Center; free (pay for services)
Controls	**Preventive = SCP / RCP**, **detective = Config**, **proactive = CloudFormation Hooks**; **mandatory (can't disable)** / strongly recommended / elective; **per registered OU**; detective only reports
Governance	**Register OU**; **enroll account** (needs `AWSControlTowerExecution`, no existing Config recorder); update landing zone -> **re-register OUs**; **drift** -> repair / re-register
Vending	**Account Factory** (Service Catalog product, `ProvisionProduct`, VPC settings); **AFC blueprints** (simplest); **AFT** (Terraform GitOps, 4 repos, `aft-invoke-customizations`); **CfCT** (manifest.yaml, StackSets + SCPs, `CreateManagedAccount` trigger)
Events	`CreateManagedAccount` lifecycle event (after baseline) -> EventBridge

Part 15 - AWS Service Catalog

Topic	Remember
Model	**Products** (CFN / Terraform, versions) in **portfolios** (access, constraints, TagOptions, sharing) -> **self-service** provisioned products
Launch constraint	**Role SC assumes** (trust `servicecatalog.amazonaws.com`) -> users need only SC permissions; **one per product per portfolio**; `LocalRoleName` for shared portfolios; admin needs `iam:PassRole`
Other constraints	**Template** (Rules limit parameters), **notification** (SNS), **StackSet**, **tag update**
TagOptions	Pick from allowed values, single value auto; SC launches only
Sharing	Account (import, can't edit products) or **Organizations** (OU, delegated admin, share principals + TagOptions) = hub and spoke
Ops	**Service actions** (SSM docs), repo sync / CodePipeline action for versions, deprecate versions, plans, `ImportAsProvisionedProduct`; Account Factory is an SC product

Part 16 - AWS Config

Topic	Remember
Recording	**Regional** recorder of configuration items; **global resources in one Region**; daily recording / exclusions for cost; timeline + CloudTrail link; **advanced queries**
Rules	**Managed** (`restricted-ssh`, `required-tags`, `approved-amis-by-id`, `cloudtrail-enabled`), **custom Lambda** (`PutEvaluations`, invoke permission), **custom Guard**; change-triggered or **periodic**; **detective** (proactive via `StartResourceEvaluation` / Hooks); **rules don't prevent**
Remediation	**SSM Automation**, manual or **automatic**, `RESOURCE_ID`, **AutomationAssumeRole**, retries, exceptions; EventBridge `Config Rules Compliance Change` for filtered alerts
Conformance packs	YAML rules + remediation (PCI, CIS, NIST); **organization packs / org rules** from management or **Config delegated admin**; members can't change
Aggregators	**Read-only** multi-account / Region view; individual = authorization; **organization = new accounts automatic**
Integrations	**Security Hub controls and Control Tower detective controls are Config rules**

Part 17 - Tagging, tag policies, Resource Groups

Topic	Remember
Basics	**Case-sensitive**, 50 per resource, `aws:` reserved, no secrets; mandatory set (CostCenter, Owner, Environment, Application) + dictionary
Apply	CFN stack tags, **CDK** `Tags.of`, Terraform default_tags, launch template TagSpecifications, **ASG PropagateAtLaunch**, **ECS propagateTags**, TagOptions, EventBridge + Lambda auto-tag
Prevent	SCP / IAM `aws:RequestTag` **+ Null** (required), `aws:TagKeys` (allowed keys / protect), `aws:ResourceTag` **/** `aws:PrincipalTag` (ABAC)
Tag policies	**Case + allowed values**, `enforced_for`; **don't require tags** (untagged = compliant); `@@assign` / `@@append` / `@@remove`; report from management account -> S3 us-east-1
Detect / cost	**Config** `required-tags` + aggregator; **Tag Editor** bulk edit; **cost allocation tags activated in management account**; **Cost Categories**
Resource Groups	**Tag-based (dynamic)** / stack-based; **regional**; SSM targets, Application Insights

Part 18 - AWS RAM

Topic	Remember
Model	Resource share = resources + principals (accounts, org, OUs, roles) + **managed permissions** (AWS or **customer managed**); **consumer IAM still required**
Org sharing	Enabled from **management account** -> no invitations, OU shares include **new accounts**; external = invitation
Shareable	**Subnets**, **TGW**, **Resolver rules**, **prefix lists**, Network Firewall, IPAM, License Manager, Capacity Reservations, Aurora (clone), Private CA, advanced SSM parameters; **not** S3 / KMS (resource policies), AMIs (launch permissions)
VPC sharing	**Same org**, no default subnets, owner = routes / NACLs / NAT, participants = own resources + SGs (no re-share, no owner default SG), **AZ IDs**, owner pays NAT
Governance	SCP `ram:RequestedAllowsExternalPrincipals`, `ram:RequestedResourceType`; **Access Analyzer**

Part 19 - Extras: OpsWorks, automation tools, Git sync, IaC generator

Topic	Remember
OpsWorks	Managed Chef / Puppet (Stacks, Chef Automate, Puppet Enterprise) - **end of life (2024)**; Stacks lifecycle Setup / **Configure** (all instances when one joins / leaves) / Deploy / Undeploy / Shutdown; new designs = **SSM State Manager** + `AWS-ApplyChefRecipes` / `AWS-ApplyAnsiblePlaybooks`
EventBridge Scheduler	Cron / rate / one-time, time zones, **universal targets** (call AWS APIs directly, no Lambda), retries, DLQ
Lambda + SDK	Execution role, **15-minute** limit, pagination, **exponential backoff**, idempotent; cross-account `sts:AssumeRole` into target roles
Step Functions	**Standard** (1 year, exactly-once) vs **Express** (5 min); SDK integrations, `.sync`, `.waitForTaskToken` (approval), **Distributed Map**, Retry / Catch
Tool choice	Schedule = Scheduler; short code = Lambda; long / multi-step = Step Functions; ops runbook + approval + rate control = **SSM Automation**
CloudFormation Git sync	Stack linked to Git branch (CodeConnections + deployment file) - **updates on push**; no test / approval stages (use CodePipeline)
IaC generator	**Scan** existing resources -> **generate template** -> **import** into stack; `cdk migrate --from-scan`

Part 20 - Domain 2 keyword list (question words to answer)

CloudFormation

The question says	Answer
Latest AMI without editing the template	`AWS::SSM::Parameter::Value<AWS::EC2::Image::Id>` public parameter
Secret in template	Dynamic reference `{{resolve:secretsmanager:...}}`
Value per Region / environment	Mappings + `FindInMap`
Create resource only in prod	Conditions
Share VPC ID with other stacks, independent lifecycle	Outputs Export + `Fn::ImportValue`
Can't delete stack output	Export in use by another stack
Reusable component deployed together / > 500 resources	Nested stacks
Reusable bundle in same stack	Module
Preview replacement before update	Change set
Someone changed resources manually	Drift detection
Protect DB from replacement during updates	Stack policy (+ UpdateReplacePolicy)
Prevent stack deletion	Termination protection
Keep data when stack / resource deleted	DeletionPolicy Retain / Snapshot
Old resource on replacement	UpdateReplacePolicy
Users can't create IAM / EC2 directly	CloudFormation service role + iam:PassRole
Configure instance from metadata	cfn-init (configSets)
Wait until app installed	CreationPolicy + cfn-signal
Wait for external process	WaitCondition + handle
Apply metadata changes to running instances	cfn-hup
ASG rolling update waiting for signals	UpdatePolicy AutoScalingRollingUpdate WaitOnResourceSignals
Replace whole ASG, quick rollback	AutoScalingReplacingUpdate WillReplace
Timed out waiting for signals	cfn-signal not sent / no outbound access
Unsupported resource / call API during deploy	Custom resource (Lambda)
Custom resource hangs 1 hour	No response to pre-signed URL
Loops / shorthand / snippets	LanguageExtensions ForEach / macro / AWS::Include
Requires CAPABILITY_AUTO_EXPAND	Macro or SAM transform
Requires CAPABILITY_NAMED_IAM	Named IAM resources
Third-party resource type	Activate public Registry extension
Block non-compliant resources before creation	CloudFormation Hooks
ROLLBACK_COMPLETE, can't update	Delete and recreate (REPLACE_ON_FAILURE)
UPDATE_ROLLBACK_FAILED	continue-update-rollback (resources-to-skip)
Roll back if alarm fires after deploy	Rollback triggers (5 alarms, 180 min)
Keep failed resources to debug	Disable rollback / DO_NOTHING
Stop a running update	cancel-update-stack
Root cause of failure	First FAILED event / CloudTrail

StackSets, CDK, SAM

The question says	Answer
Same stack in many accounts and Regions	StackSets
New accounts in OU get it automatically	Service-managed + auto-deployment
Accounts outside the org / management account	Self-managed StackSet
Delegated admin 'StackSet not found'	`--call-as DELEGATED_ADMIN`
New Region for existing StackSet	create-stack-instances
Exclude one account in OU	AccountFilterType DIFFERENCE
Stop on first failure, one Region at a time	FailureTolerance 0 + SEQUENTIAL
OperationInProgressException	Managed execution
IAM role fails in 2nd Region	Global resource, deploy in one Region
Infra in familiar programming language with tests	AWS CDK
Bootstrap version not found	cdk bootstrap account + Region
CDK pipeline can't deploy to other account	`--trust` + crossAccountKeys
Renamed construct, table replaced	Logical ID changed, overrideLogicalId
fromLookup fails	Set env account + Region
Synth differs in pipeline	Commit cdk.context.json
Pipeline updates itself	CDK Pipelines self-mutation
Parallel Regions in CDK pipeline	addWave
Serverless IaC + local testing	AWS SAM
Test API locally	sam local start-api
SDK tests against local Lambda	sam local start-lambda
Native dependency error in Lambda	sam build --use-container
Lambda canary with auto rollback	AutoPublishAlias + DeploymentPreference + Alarms
Fast dev code sync	sam sync --watch / cdk watch (dev only)

Systems Manager and AppConfig

The question says	Answer
Instance not managed	Agent + instance profile + endpoints
Private subnet, no NAT	Interface endpoints ssm / ssmmessages / ec2messages
SSM for all EC2 without instance profiles	DHMC
On-prem servers	Hybrid activation
Script on many instances now, rate limited	Run Command concurrency + error threshold
Always configured incl. new instances	State Manager association
Different patch delays prod / dev	Custom baselines + Patch Group + maintenance windows
Report missing patches only	AWS-RunPatchBaseline Scan
Org-wide patching	Patch policies
Installed software across accounts with SQL	Inventory + Resource Data Sync + Athena
No bastion, audit commands	Session Manager + logging (S3 / CW, KMS)
Laptop to private RDS	Port forwarding to remote host
SecureString AccessDenied	kms:Decrypt

The question says	Answer
Value over 4 KB / expiry notice	Advanced parameter + policies
Automatic secret rotation	Secrets Manager
Multi-step ops with approval	Automation runbook aws:approve
Lost EC2 key pair	AWSSupport-ResetAccess
Custom agent everywhere	Distributor + State Manager
RDP without 3389	Fleet Manager
Block changes during freeze	Change Calendar
Change behaviour without redeploying	AppConfig feature flag
Validate config before deploy	AppConfig JSON Schema / Lambda validator
Roll back config on alarm	AppConfig environment monitors + bake time
Lambda reads config cheaply	AppConfig Lambda extension localhost:2772

Organizations, Control Tower, Service Catalog

The question says	Answer
Prevent action in all member accounts	SCP deny
Management account not blocked	SCPs don't apply there
Region lock breaks IAM / CloudFront	NotAction for global services
Prevent leaving org	SCP deny LeaveOrganization
Invited account role switch fails	Create OrganizationAccountAccessRole
Org accounts read bucket, no ID list	aws:PrincipalOrgID
Block outside access despite bad policies	RCP
EC2 baseline that survives new APIs	Declarative policy
Keep RIs in one account	Turn off RI / SP sharing
Org security without management account	Delegated administrator
Members can't stop logging	Organization trail
React to new account	EventBridge us-east-1 / CreateManagedAccount
Best-practice multi-account quickly	Control Tower landing zone
Block / report / before CFN	Preventive / detective / proactive controls
Can't disable control	Mandatory
Existing OU under governance	Register OU
Enroll fails	AWSControlTowerExecution role / Config recorder
After landing zone update	Re-register OUs
Account moved outside CT	Drift, update / re-enroll
Terraform GitOps vending	AFT
CFN + SCPs auto for new accounts	CfCT
Simplest console customization	AFC blueprints
Self-service approved stacks, no service permissions	Service Catalog + launch constraint
Limit instance types users pick	Template constraint
One launch constraint for all spokes	LocalRoleName
Restart without EC2 permissions	Service actions

The question says	Answer
Share portfolio to OU	Organizations sharing + delegated admin

Config, tagging, RAM

The question says	Answer
What it looked like / who changed it	Config timeline / CloudTrail
Detect open SSH	Managed rule restricted-ssh
Custom check with external data	Custom Lambda rule
Simple check, no Lambda	Guard rule
Daily time-based check	Periodic trigger
Auto-fix non-compliance	Config remediation + SSM Automation
Remediation fails	AutomationAssumeRole
Email only on NON_COMPLIANT	EventBridge filter
Same rule set in every account, protected	Organization conformance pack
Central compliance view incl. new accounts	Organization aggregator
Aggregator fixes resources?	No, read-only
Security Hub no findings	Enable Config
Require tag at creation	SCP aws:RequestTag Null
Standard tag case / values org-wide	Tag policy
Untagged launch allowed despite tag policy	Tag policies don't require tags
Tags missing in Cost Explorer	Activate cost allocation tags (management)
Team-scoped access at scale	ABAC PrincipalTag = ResourceTag
Run command on app's resources incl. new	Tag-based resource group
App accounts launch into central VPC	VPC sharing (RAM)
No invitations, new accounts automatic	RAM org sharing to OU
Org-wide on-prem DNS forwarding	Share Resolver rules
Same CIDRs in all security groups	Shared prefix list
Block sharing outside org	SCP ram:RequestedAllowsExternalPrincipals

Extras

The question says	Answer
New Chef / Ansible config management	SSM State Manager (OpsWorks is end of life)
OpsWorks event when instances join / leave	Configure lifecycle event
Scheduled API call without code	EventBridge Scheduler universal target
Automation job longer than 15 minutes	Step Functions Standard
Human approval in a workflow	Step Functions .waitForTaskToken / SSM aws:approve
Automation across many accounts	AssumeRole into each target account
SDK throttling / only first page	Exponential backoff + pagination
Millions of S3 objects in parallel / start job and wait	Distributed Map / .sync
Stack updates on Git push, no pipeline	CloudFormation Git sync
Manually created resources into IaC	IaC generator + import / cdk migrate --from-scan

Part 21 - Numbers to remember

Item	Number
Domain 2 weight	**17%**
CloudFormation resources per stack	**500**
Template body / S3	51,200 bytes direct, **1 MB** from S3
Parameters / outputs / mappings	200 each
CreationPolicy / WaitCondition timeout	max **12 hours**
Custom resource default wait	**1 hour**
cfn-hup default interval	**15 minutes**
Rollback triggers	**5** alarms, **0 to 180** minutes
StackSets delegated admins	**5**
SCPs per target / size	**5** / **5,120** characters
OU nesting depth	**5**
Tags per resource / key / value	**50** / 128 / 256
Config `required-tags` keys	**6**
Config periodic frequencies	1, 3, 6, 12, 24 hours
Parameter Store Standard / Advanced	**4 KB**, 10,000, free / **8 KB**, 100,000, paid
Run Command console output	**2,500** characters
AppConfig agent port	**2772**
SAM local start-api / start-lambda	**3000** / **3001**
Lambda max timeout / Express workflow / Standard workflow	**15 min** / **5 min** / **1 year**
Service Catalog launch constraints	**1** per product per portfolio
RAM invitation	about **7 days**
Cost allocation tag activation	up to **24 hours**

Part 22 - Classic wrong answers (distractors)

Distractor	Why it's wrong
Tag policy to require tags	Tag policies don't require tags - use SCP `aws:RequestTag`
Config rule to prevent changes	Config is detective - use SCP / Hooks
Aggregator to deploy or remediate	Aggregators are read-only
SCP to restrict the management account	SCPs never apply there
SCP to grant permissions	SCPs only limit
Service-managed StackSet to the management account	Never deploys there
Auto-deployment adds new Regions	Only reacts to OU membership
Parameter Store for automatic rotation	Rotation = Secrets Manager
Run Command for future instances	That's State Manager
Patch baseline alone patches	Needs AWS-RunPatchBaseline task
Edit the AWS predefined patch baseline	Create a custom one
CloudFormation update on ROLLBACK_COMPLETE	Delete and recreate
update-stack on UPDATE_ROLLBACK_FAILED	continue-update-rollback
NoEcho encrypts secrets	It only masks - use dynamic references
Exports across Regions / accounts	Same account + Region only (use SSM parameters)
cdk deploy --hotswap in production	Dev only, causes drift
sam local enforces IAM	Uses your local credentials
Lambda env vars for feature flags	Needs a redeploy - use AppConfig
Control Tower detective controls auto-fix	Only report - add remediation
Control Tower enables GuardDuty	No - Audit as delegated admin
Service Catalog to push a baseline to all accounts	Pull model - use StackSets / CfCT
Share S3 / KMS with RAM	Resource policies
VPC sharing with external accounts	Same organization only
Organizations CreateAccount event for CT automation	Use CreateManagedAccount
OpsWorks for a new config management design	End of life - Systems Manager
Lambda for a multi-hour job	15-minute limit - Step Functions
Git sync when approvals are required	No stages - CodePipeline